# The Opposing Effects of Complexity and Information Content on Uncertainty Dynamics: Evidence from 10-K Filings *

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**Abstract**

We evaluate the impact of complexity and information content of 10-K filings on uncertainty dynamics following the filings. We have three main findings. First, the option implied volatility on average increases in the first four weeks after the filings, followed by a net decrease in the subsequent six weeks. Second, this hump-shaped volatility dynamics is more pronounced for firms with larger 10-K file sizes. Third, we provide a novel decomposition of 10-K file size based on the individual sections' disclosure amount and topic analysis, and find that the discussions on topics in the Risk Factors section mainly capture the complexity aspect whereas the discussions on topics in the Managerial Discussion and Analysis (MD&A) section mainly capture the information content aspect of the 10-K filings. Our findings highlight the importance of the timing for understanding the opposing effects of complexity and information content on asset prices.

Keywords: Learning; Complexity; Information; Volatility dynamics; Textual analysis
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# 1 Introduction

The fundamental goal of annual reports (10-Ks) is to provide information to investors. In principle, the information content in annual reports should lead to a resolution of uncertainty, with more information content leading to less uncertainty. However, recent empirical studies provide evidence that seems to be inconsistent with the fundamental goal of 10-K filings. For instance, Loughran and McDonald (2014) (henceforth LMD) document that a larger 10-K filing is associated with higher stock return volatility in a short horizon following the filing, suggesting that the size of 10-K could be associated with the complexity of the annual report.[1] In this paper, we study the dynamics of volatility following 10-K filings and reconcile the seemingly contradictory empirical evidence of LMD with the fundamental information role of 10-K filings.

Motivated by the idea that investors take time to learn about important corporate attributes (e.g., Pan, Wang, and Weisbach (2014)), we conduct a comprehensive analysis of the uncertainty dynamics in the event window following 10-K filings. We use option implied volatility as the baseline measure of investors' perceived uncertainty. Following LMD, we consider the file size as the main cross-sectional characteristic of a 10-K report. We show that the 10-K file size not only captures the complexity of 10-K filings, but also is a good proxy for the amount of information content, consistent with its fundamental goal and aggregate nature.

Our main empirical findings can be summarized as follows. First, we document a hump-shaped volatility dynamics following 10-K filings as in Figure 1. Specifically, an average firm experiences a 0.4% increase in volatility in the first two to four weeks (short horizon) after 10-K filings, followed by a 2.6% decrease in volatility in the subsequent six weeks (long horizon), leading to a net reduction of 2.2% relative to the volatility level around the filing date. Second, we find that this hump-shaped pattern is more pronounced for firms with larger (hence more complex but potentially more informative) 10-K file size (Figure 2). Consistent with LMD, we find that firms with a larger 10-K experience a bigger increase in volatility than firms with a smaller 10-K in the first two to four weeks after the 10-K filing. More importantly, we find that firms with a larger 10-K also experience a sharper decrease in volatility in the subsequent six weeks. All else being equal, a one standard

[1] More specifically, Loughran and McDonald (2014) document that a larger 10-K report is associated with higher stock return volatility in the [6,28] trading day period following the 10-K filings. As a result, LMD propose the 10-K file size as an easy-to-construct yet effective proxy for the complexity of 10-K report.



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deviation increase in log 10-K file size is associated with 0.31% increase in short-run volatility, but 0.62% net decrease in volatility in the long horizon. The patterns are similar for alternative measures of uncertainty, such as ambiguity, analyst forecast dispersion, and the volatilities constructed using realized stock returns.

To sharpen the identification of the mechanism at play and better understand the economic underpinnings of the 10-K file size, we conduct topic analysis on the textual content of the 10-K filings. Although both the Risk Factors and the MD&A sections' disclosure amount (measured by the aggregate topic score in each section) are positively correlated with the 10-K file size, the correlation between hard information (measured by percentage of numbers in 10-Ks) and disclosure amount is negative for Risk Factors and positive for MD&A. Furthermore, while the aggregate topic score of Risk Factors is positively associated with the short-run volatility dynamics and not significantly related to the long-run dynamics, that of MD&A is negatively associated with the long-run volatility dynamics and not related to the short-run dynamics. These results suggest that, on the one hand, text-heavy disclosure of Risk Factors primarily captures the complexity component of 10-K file size, leading to an increase of uncertainty in the short run. On the other hand, the information content component of 10-K file size is driven by more of hard information disclosure in MD&A, leading to a reduction of uncertainty in the long run.

Additionally, we find that the percentage of numbers in 10-Ks itself is an effective normalized proxy for 10-K information content. It correlates positively with the 10-K file size and negatively with both the short-run and long-run volatility changes. We also test an alternative 10-K file size measure that better captures the amount of "new" information (through the dissimilarity of word usage) while minimizing the complexity aspect in the report. We find that this measure has a significant predictive power for the reduction in volatility in the long run but has a negligible effect on the short-run volatility dynamics.

The impact of the 10-K filings on the volatility dynamics also varies with the transparency of firm's information environment. Firms with low analyst coverage or low institutional ownership have a stronger relation between 10-K file size and the short-run increase in volatility. In contrast, firms with high analyst coverage or high institutional ownership have a stronger relation between 10-K file size and the long-run decrease in volatility. These results suggest that less transparent information environment impedes information incorporation and investors perceive higher



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uncertainty in the short run. However, when 10-K file size is large, the ultimate incorporation of information is more thorough and subsequence uncertainty resolution is more complete, giving rise to a more reduction in the long-run implied volatility.

The magnitudes of our main results on the cross-sectional relation between 10-K file size and volatility dynamics are comparable to those reported in the related studies (e.g., Loughran and McDonald (2014); Pan, Wang, and Weisbach (2014)). To illustrate the economic significance of the hump-shaped volatility dynamics, we explore an option strategy that involves shorting straddles between two-to-four weeks and eight-to-ten weeks following the 10-K filings. We find that for firms with large 10-Ks, this strategy delivers up to 16% higher annualized returns than for firms with small 10-Ks.[2]

We perform additional analyses to establish the robustness of our empirical findings. For example, the selection of short-run and long-run horizons may be arbitrary in our main analysis. To alleviate this concern, we use equally-spaced expanding windows to measure the volatility dynamics and find that the patterns of findings are consistent with what we observe in Figure 2. Another important concern is the impact of confounding information in our event study setting. We control for several sources of information including the corporate material information disclosure in 8-Ks, a comprehensive news coverage based on RavenPack dataset, specific news related to analysts ratings and forecasts, and a measure of firm complexity (number of business segments). The results are robust to controlling for all these additional variables. In addition, we use different approaches to investigate to what extent our results might be related to subsequent earnings announcements. We find the impact of earnings announcements is likely to be minimal.

Taken together, our empirical analyses uncover the information role of 10-K filings at the longer horizons of four to ten weeks following 10-K disclosures. Consistent with the fundamental goal of annual reports, a larger 10-K filing carries more information, although it takes time for investors to process and digest it. Coupled with the finding by LMD, our results indicate that the dual roles of information (i.e., complexity and information content) in complex but informative signals, such as a firm's 10-K report, can be inferred from the dynamics of return volatility as investors learn

[2]From a theoretical point of view, the Black-Scholes options pricing model implies a smooth, monotonically increasing, and convex relation between the underlying asset return volatility and option prices for both call and put options. Given that the implied volatility at the firm level is highly persistent, options price can be locally approximated as a linear function of volatility, so predicting volatility changes can also be considered as predicting option returns.



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through time. Our findings thus offer empirical guidance for theoretical research on learning and asset pricing, especially in settings where the informative signal is difficult to process.[3]

*Background on 10-K and Theoretical Foundations*

A Form 10-K is an annual report required by the U.S. Securities and Exchange Commission (SEC), that gives a comprehensive summary of a firm's financial and operational performance. Besides the key numbers such as earnings, dividends, and sales growth that have been disclosed to the public in earnings announcements, 10-K filings typically run for hundreds of pages and also include a lot of detailed information regarding the risk factors, financial conditions, operational results, managerial compensations, as well as projections on the firm's future performance. This large amount of additional information about the company is important and valuable to investors. For example, Griffin (2003) documents that the average absolute value of excess return is reliably greater on the filing date and the magnitude of investor response has increased over time. You and Zhang (2009) present empirical evidence that the trading volumes and the average absolute stock returns during a 4-day period starting from the filing date are abnormally high. In addition, Cohen, Malloy, and Nguyen (2020) suggest that there is a large amount of rich information that is being hidden in the 10-Ks. They find that investors miss subtle changes in 10-K languages perhaps due to their increased complexity and length but these changes are important signals that can generate a long-short stock return spread of over 22% per year in the future. There are also many other papers surveyed by Loughran and McDonald (2016) that examine the information of 10-Ks in various settings.

If 10-K filing is truly *informative*, it should help resolve some uncertainty regarding the future prospects of the company. This information content aspect of 10-K filings is consistent with standard Bayesian learning models with information signals (see, e.g., Pastor and Veronesi (2009)). However, due to the large average 10-K file size (approximately 1.37 megabytes in our sample), it is unlikely for investors to digest all of the information immediately after the release. Instead, it may take some time, perhaps a couple of weeks or even months, for investors to process the

[3]Papers that examine the impact of learning on stock volatilities include Timmermann (1993), Brennan and Xia (2001), Pastor and Veronesi (2003), among many others. Interested readers should refer to Pastor and Veronesi (2009) for an excellent review of how learning affects various aspects of financial markets. Peng and Xiong (2003) study the effect of analysts' cost of effort on long-run volatility persistence. Another interesting line of research focuses on feedback learning from prices (e.g., Chen, Goldstein, and Jiang (2007), Edmans, Goldstein, and Jiang (2012), Edmans, Goldstein, and Jiang (2015), Goldstein, Schneemeier, and Yang (2021), Liu, Goldstein, and Yang (2021)).



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10-K filing and incorporate it into the asset prices. For instance, Lehavy, Li, and Merkley (2011) use the amount of time it takes analysts to issue reports following 10-K filings (or "analyst report duration") to measure the cost or effort that analysts bear in following firms, and document that the average analyst report duration is about 18 days.

Combining the above considerations, we discuss our findings on the dynamics of uncertainty following 10-K filings in the theoretical context of investor learning. In the short run, the complexity of 10-K may cause an increase of the uncertainty of a firm's information environment and therefore lead to a higher level of volatility. In a relatively long horizon, however, as the information is gradually digested by investors, the 10-K information can help resolve uncertainty. To the extent that 10-K report contains informative signals, it should reduce the uncertainty to a level even lower than the filing-week level.

Although the initial increase in uncertainty following 10-K filings cannot be easily rationalized by standard Bayesian learning models with informative signals, it may be consistent with some theoretical models under alternative assumptions. For example, when private information acquisition is endogenized (Diamond (1985)), Goldstein and Yang (2017) show that at intermediate range of public information precision, the volatility could increase when the public information precision increases. This is because of the crowding out of private information trigged by the increased public information precision. Alternatively, costly information processing may also change the effect of 10-K disclosures on uncertainty, especially in the short run. In a rational expectations equilibrium model, Mondria, Vives, and Yang (2021) show that when investors are boundedly rational and do not fully understand how to extract information from asset prices, the processing cost can inject a noise term into the price system that leads to excess return volatility and trading volume.[4]

*Related Literature*

Our paper is related to several lines of research. First, our study contributes to the emerging studies on how investor learning over time about important corporate attributes affects the perceived riskiness of corporate securities. For example, Pan, Wang, and Weisbach (2014) study the

[4]In addition, in theoretical settings where investors are ambiguity averse, investors might not know which information to process due to the large amount of complex information (e.g., Epstein and Schneider (2007)). For example, Epstein and Schneider (2007) show that new complex information may make investors less confident and more ambiguous and perceive higher uncertainty. Even in the absence of ambiguity aversion or information processing cost, Kondor (2012) shows that if investors have different trading horizons and heterogeneous private information, public announcements can lead to increases in the short-run volatility.



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impact of learning about managerial ability on volatility dynamics. Our paper differs in the objective of learning (10-K information) and horizon of investor learning as we examine the volatility dynamics in a much shorter window of 10-weeks following 10-K filings as opposed to a much longer horizon of 3 years following CEO turnovers in their paper. Other related studies in this area include Pan, Wang, and Weisbach (2015), who investigate the impact of learning about management risk on cost of borrowing, and Giglio and Shue (2014), who study the impact of investor learning on M&A outcomes. We contribute to this line of research by demonstrating the novel impact of learning on volatility dynamics in the important setting of 10-K filings, and in particular uncovering the fundamental information role of 10-K filings.

Second, our paper contributes to the large literature on the informativeness and complexity aspects of corporate information by showing two important sections of 10-Ks playing different roles with Risk Factors mainly capturing complexity and MD&A mainly capturing informativeness.[5] Most studies on information disclosures consider one aspect at a time while ignoring the other. Our main contribution to this literature is to unify these two lines of research by highlighting the importance of timing in understanding the dual roles of information releases. In particular, our paper adds to the literature that aims at understanding the informativeness of 10-K filings using textual analysis (see Loughran and McDonald (2016) for a recent survey of the literature). For example, Cohen, Malloy, and Nguyen (2020) show that the future returns of 10-K/Q changers and non-changers are significantly different, highlighting the informativeness of the corporate filings. Our paper differs in several dimensions. Our paper focuses on the uncertainty dynamics in the 10-week horizon following 10-K filings whereas their paper focuses on the price drift in a longer time period. Our measures of information content complement theirs and our results are robust to controlling for their measure. In addition, our analysis reveals the importance of hard information in 10-K filings, complementing some other related papers in different settings.[6] Another related

[5] See Leuz and Wysocki (2016) and Loughran and McDonald (2016) for two recent surveys of the disclosure literature. For an incomplete list of papers on the informativeness of disclosures, see for example, McNichols and Manegold (1983), Lang and Lundholm (1996), Botosan (1997), Botosan and Plumlee (2002), Kothari, Li, and Short (2009), Berger, Chen, and Li (2012). On the complexity aspect, see for example, Li (2008), You and Zhang (2009), Lehavy, Li, and Merkley (2011), Dougal, Engelberg, Garca, and Parsons (2012), Lawrence (2013), De Franco, Hope, Vyas, and Zhou (2013), Loughran and McDonald (2014).

[6] For example, Zhang, Zhao, and Zhao (2020) find that the more word usage (as opposed to numbers) in the offering documents indicates a higher risk in the residual mortgage backed securities (RMBS) in the recent financial crisis. Zhou (2020) finds that more usage of numbers (as opposed to words) in corporate earnings conference calls reflects positive information and induces positive price reactions.



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paper is Kogan, Levin, Routledge, Sagi, and Smith (2009), which find that words in 10-K filings help predict 12 month ahead stock return volatility (via support vector regressions) more accurately, although their goal is not to analyze the direction of how the words affect the future volatilities.

# 2 Main empirical findings

This section reports the main empirical findings. In Section 2.1 we describe the data. In Section 2.2 we show firms' option implied volatility dynamics following 10-K filings, and how the volatility dynamics vary with 10-K file sizes. In Section 2.3 we report the results using several alternative measures of uncertainty dynamics.

## 2.1 Data

Our sample is constructed using several data sources. We obtain the option data from OptionMetrics, the stock data from CRSP and TAQ, the accounting data from Compustat, the analyst forecast data from IBES, and the news data from RavenPack. We also obtain the 10-K file size data from Tim Loughran and Bill McDonald and the 10-K individual section data from SeekEdgar.[7]

Our goal is to analyze the impact of 10-K filings on the evolution of investor's perceived uncertainty. Therefore, we use the option implied volatility (IV) as the benchmark measure of firm-level uncertainty due to IV's forward looking nature.[8] However, our results are robust to using several alternative measures of uncertainty.

In order to cover the horizon of investor learning in the period of 10 weeks after 10-K filings, we use options with relatively longer maturities to construct our main measure of volatility dynamics. In particular, we include options with 62 to 213 days (or 2 to 7 months) to maturity and positive trading volumes for constructing the firm-level implied volatilities. We take the mid-quote value as a fair reflection of the option price. We also require the option price to satisfy the arbitrage

[7] We thank Tim Loughran and Bill McDonald for making the 10-K filing data available online at http://www3.nd.edu/ mcdonald/Data/LoughranMcDonald_10-X_FileSize.csv.

[8] There is a line of research suggesting that this uncertainty measure is superior to alternative measures that are based on realized stock returns in capturing investor's perceived uncertainty. For example, this could be due to the forward-looking nature of the implied volatility and that it is less affected by large realized first moment shocks on the underlying equity. For an incomplete list of contributions in this literature, see for example, Day and Lewis (1992), Canina and Figlewski (1993), Lamoureux and Lastrapes (1993), Christensen and Prabhala (1998), Fleming (1998), Blair, Poon, and Taylor (2001), Jiang and Tian (2005), Busch, Christensen, and Nielsen (2011), Chang, Christoffersen, Jacobs, and Vainberg (2012). Poon and Granger (2003) also provide a nice review of relevant studies.



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bound. For call options, the arbitrage bound is max(0, stock price−strike price)<mid quote<stock price. For put options, the arbitrage bound is 0<mid quote<strike price. The moneyness of an option – the ratio between the stock and strike prices – is required to be between 0.8 and 1.2. After screening the option data, we calculate the firm-level daily implied volatility as the simple average of the implied volatilities of all remaining options on that firm on each trading day.

Our main characteristic regarding 10-K filings is the file size from LMD due to its potential capability of capturing the dual roles of information: complexity and information content. LMD suggest that if firms want to obscure mandated earnings-relevant information, they are more likely to bury the results in longer documents. Indeed, they find a positive association between 10-K file size and future uncertainty in a horizon of one month following 10-K filings. However, given the fundamental goal of financial disclosures, larger 10-Ks may also contain more information content despite its difficulty of comprehension in a short period of time.

We follow LMD to control for several other firm-level characteristics in our analysis, including the firm's market cap, book-to-market equity ratio, the pre-filing CAPM alpha in the window of [-252,-6] days before the 10-K filing, and the absolute value of filing date excess return. As in LMD, we further impose the following restrictions on the underlying stocks in our sample: common shares (SHRCD to be 10 or 11) and stock price $\geq$ \$3.[9]

In the online appendix, we report the summary statistics of the key variables for the benchmark sample from 1996 to 2012. The logarithm of 10-K file size (Fsize) has a mean of 0.317 and a standard deviation of 1.265, and it varies substantially across firm-year observations. For instance, the 10-K file size for the 75th percentile is 2.75 megabytes, more than five times larger than that for the 25th percentile. Bigger firms have larger 10-K reports on average. The correlation between Fsize and firm size is 26.5%, suggesting that file size has a large variation that can be driven by the amount of information content independent of firm size.

## 2.2 Volatility dynamics following 10-K filings

Figure 1 provides baseline evidence for the volatility dynamics following the 10-K filings. To make the volatility dynamics comparable across firms, we benchmark all the volatility observations

[9] Our results are essentially the same when we use the alternative price cutoff of \$5 or \$1, as reported in the online appendix.



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following 10-K filings to the filing-week volatility and calculate the percentage changes.[10] In other words, the volatility change in the $i^{th}$ week from the filing week (week 0) is defined as $100 \times \ln[IV(i)/IV(0)]$, where $IV(i)$ is the average volatility in week $i$, and $IV(0)$ is the average volatility in the filing-week. We define the filing-week as the week prior to and including the 10-K filing day. Figure 1 shows a hump shape in the volatility dynamics – a mild increase in weeks 2–4 (with a peak at around week 2), followed by a rapid decline in weeks 8–10.[11]

[Insert Figure 1 here]

Figure 2 shows our main results on the impact of 10-K file size on the shape of volatility dynamics. In each calendar year, firms are sorted into quintile based on 10-K file size. The volatilities are then averaged within a quintile and across calendar years, and the resulting volatility dynamics are plotted in the figure. The left plot shows that while both firms with large 10-Ks and firms with small 10-Ks display hump-shaped volatility dynamics, the pattern is more pronounced for large 10-K firms. Following 10-K filings, the cumulative percentage change in volatility is about 1.5% (0.6%) for firms with large (small) 10-Ks by weeks 2–4, but becomes −4.5% (−1.9%) by week 9. This pattern can also be seen in the right panel of this figure, in which the difference in the volatility dynamics between the top and bottom quintile portfolios displays a hump shape.

[Insert Figure 2 here]

We quantitatively evaluate the patterns in the figures using panel regressions. We define diffs as the average percentage change in the implied volatility in the short run (2-4 weeks) from the 10-K filing and diffl as the average percentage change in implied volatility in the long run (8-10

[10] Another important reason is that the level of volatility is highly persistent and it is typically modelled using GARCH models with time series data. In our OLS panel regression setting, the low persistence of volatility changes is less subject to this econometric issue. In our regressions we control for the year fixed effects and cluster standard errors by firm to deal with the issue.

[11] The increase in implied volatility after week 9 can be due to the subsequent quarterly earnings announcement (e.g., Patell and Wolfson (1979) and Gao, Xing, and Zhang (2018)). According to the SEC, Form 10-K had to be filed within 90 days after the end of the company's fiscal year. However, in September 2002, the SEC approved a Final Rule that changed the deadlines to 75 days for Form 10-K for "accelerated filers" that have a public float of at least $75 million. In December 2005, the SEC created a third category of "large accelerated filers", accelerated filers with a public float of over $700 million. Beginning with the fiscal year ending on or after December 15, 2006, the deadline for "large accelerated filers" is 60 days. The average and median gap between 10-K filing dates and subsequent quarterly earnings announcement dates are both about 9 weeks in the most recent regulation regime since 2006; the $1^{st}$ and $3^{rd}$ quartile of this gap are 8 and 11 weeks respectively. Note the final increase should not be surprising because if the average volatility displays a hump shape in each quarter without such a final increase, volatility would have a counterfactual decreasing trend.



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weeks). We also define diffsl as the difference between diffs and diffl to capture the dynamics of uncertainty from the short run to long run. The choice of the short-run and long-run horizons is motivated by previous studies (e.g., LMD) and the observations from Figure 1.

Panel A of Table 1 reports the average diffs, diffl, and diffsl for the full sample from 1996 to 2012. Following the 10-K filing, the average implied volatility increases by 0.4% in the first 2 to 4 weeks. This increase is statistically significant, reflecting the complexity aspect of an average 10-K filing. However, as the information is gradually understood by investors, the information content aspect of 10-Ks becomes more important and starts to dominate the impact on uncertainty. Indeed, we observe a rapid decline of 2.6% in implied volatility from weeks 2–4 to weeks 8–10, resulting a 2.2% net decrease in implied volatility from the filing-week level.

Next, we focus on the impact of 10-K file size on the shape of volatility dynamics. LMD highlight 10-K file size as a proxy for disclosure complexity, not only because a larger file size requires more effort to understand, but also because firms may try to obscure mandated earnings-relevant information and bury the results in longer documents. However, the file size, as an aggregate measure of a 10-K filing, can also be a natural proxy for the amount of information. Despite the difficulty of comprehension due to complexity, a large 10-K can contain a great amount of information that, once digested, helps resolve the firm's uncertainty. Therefore, we expect a larger file size to be associated with a stronger hump-shaped volatility dynamics following 10-K fillings.

In Panel B of Table 1, we regress diffs, diffl, and diffsl, on the natural logarithm of 10-K file size with or without controlling for other firm-level and disclosure-specific characteristics, including firm size, book-to-market equity ratio, the logarithm of the implied volatility, the CAPM alpha $[-252,-6]$ days before 10-K-filings, and the absolute value of filing period return. The results without other control variables are reported in columns (1), (3), and (5) and the results with control variables are reported in columns (2), (4), and (6). Overall the results with and without other control variables are similar and the coefficient on Fsize is statistically significant at the conventional levels. To understand the economic impact of file size, we focus on columns (2), (4), and (6).[12] All else being equal, a one standard deviation increase in log 10-K file size is associated

[12]Note that the signs of some control variables such as firm size and pre-alpha in the regression of diffs (column (2)) are different from Loughran and McDonald (2014). This is because we focus on the volatility dynamics using percentage changes. We are able to replicate Loughran and McDonald (2014) if we use the volatility level as the dependent variable.



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with 0.248 × 1.265 = 0.314 percent increase in implied volatility in 2–4 weeks, but 0.487 × 1.265 = 0.616 percent decrease in implied volatility in 8–10 weeks following the 10-K filing.

[Insert Table 1 here]

The positive relation between 10-K file size and short-run volatility change is consistent with LMD, who find a similar pattern and propose file size as an effective measure for readability and complexity of financial disclosures. More importantly, our analysis on the volatility dynamics reveals the crucial role of information, i.e., information content, in the longer horizon. Our identification of this dual role of information through the time dimension therefore complements LMD who only focus on the complexity aspect of information.

### 2.3 Alternative measures of uncertainty dynamics

Although our main measure of uncertainty using option implied volatility is forward looking, there are alternative measures of uncertainty that capture different aspects of uncertainty in the information environment.

Our first alternative uncertainty measure is based on the firm-level ambiguity of the stock return distributions. Following Izhakian and Yermack (2017) and Augustin and Izhakian (2020), we estimate the degree of ambiguity based on 5-minute stock returns for each firm using intraday data from the TAQ database.[13] We measure the ambiguity for each firm week using rolling 21-day return distributions, and then calculate the average percentage change in the ambiguity 2–4 weeks (diffs) and 8–10 weeks (diffl) following the 10-K filings.

The second and third alternative measures are based on the information of analyst forecasts: analyst forecast dispersion and the absolute value of earnings surprises (SUE). Since our goal is to understand the dynamics of investors' perceived uncertainty, it is important for us to measure the dynamics both in the (relatively) short and long horizons. We first construct these measures at monthly frequency considering that analyst forecasts are much more sparse than the volatility measures. We then measure the dynamics of analyst forecast dispersion of each firm-year in three

[13] We form a grid of 160 bins from -40% to 40% in addition to the left and right tails for the daily intraday return distributions. The ambiguity measures are derived from the expected probability (and the variance) of being in a particular bin across these daily return distributions. We drop all days with less than 15 different 5-minute returns and also drop months with less than 15 intraday return distributions.



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different horizons: the percentage changes in the dispersion from the 10-K filing to one month after the filing (diffs), from the 10-K filing to up to three months after the filing (diffl), and the difference between diffs and diffl (diffsl), so that the horizons match with those used in the analysis of the volatility dynamics. While it is rather challenging to measure the dynamics of the absolute value of earnings surprises in the same way (since there is only one quarterly earnings announcement after 10-K filing), we conduct a similar analysis by estimating SUE in each month based on analysts' forecasts reported in that month.

The fourth and fifth measures are based on residual root mean squared error (RMSE) and the implied volatility from options with 30 days to maturity (30-day options) respectively. As in LMD, we measure RMSE from a market model estimated using daily observations (rolling 21 trading days) of individual firms' returns. We then calculate the percentage changes of these uncertainty variables for the short and long run in the same way as we calculate our main volatility dynamics.

[Insert Table 2 here]

Panel A of Table 2 shows the results of how ambiguity dynamics are impacted by the 10-K file size. Similar to the dynamics of implied volatility, we find that the ambiguity increases in the short-run and decreases in the long-run. This result indicates that the size of 10-K filings has differential implications on the ambiguity dynamics in the short and long horizons. The dynamics of the analysts forecast dispersion (Panel B) and the absolute value of SUE (Panel C) are also higher in the short run but lower in the long run for firms with larger 10-K filings, suggesting that the analysts' perception of uncertainty evolve in a similar pattern to the general investors. Panel D reports the analysis based on the realized stock returns (RMSE) and we again find that the hump-shaped dynamics is more pronounced for larger 10-K filings. In Panel E, we address the issues of liquidity and differential horizons for the longer maturity options used in our main regressions by focusing exclusively on 30 day options. The effect of 10-K file size on the 30-day option implied volatility dynamics is similar to our main findings in both statistical and economic significance. For example, all else being equal, a one standard deviation increase in log 10-K file size is associated with $0.507 \times 1.265 = 0.641$ percent increase in the 30-day implied volatility in the short horizon, but $0.512 \times 1.265 = 0.648$ percent decrease in the 30-day implied volatility in the long horizon.

In the online appendix, we explore alternative empirical specifications such as volatility dy-



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namics with equally-spaced window and volatilities based on higher-frequency realized returns. We find similar results as in the main analysis. Moreover, we also analyze how 10-K file size affects the short- and long-run changes in two commonly used information asymmetry measures: Amihud (2002) illiquidity and bid-ask spread, as well as an information efficiency measure: the absolute value of the centered variance ratio as in Boehmer and Kelly (2009). We find significant decreases in Amihud illiquidity, bid-ask spread, and absolute variance ratio in the long run for firms with larger 10-Ks, while these measures do not increase significantly with 10-K file size in the short-run.[14] Overall, these findings suggest that investors do not fully process the 10-K information in a short period of time and the impact of 10-K file size on the uncertainty dynamics is robust to the alternative measures of uncertainty.

# 3 Text-Based Separation of Information Content and Complexity

Although the file size is an easy-to-construct measure of 10-K filings, in this section, we analyze the content of 10-K filings to enrich the economic underpinnings of 10-K file size. In particular, we construct text-based measures to capture the complexity and information content of 10-K filings separately. Our approach is motivated by the idea that information disclosure is multifaceted, and it is important to understand what is being disclosed, utilizing topic analysis.

The first step of our topic analysis is to single out two important sections of 10-K filings, the Risk Factors and the MD&A sections, that are critical sources of firm valuation signal, yet could convey firm risks in different ways.[15] For example, the Risk Factors section tends to cover various risk factors with heavy text usage and it generally focuses on the risks themselves, not how the company addresses those risks.[16] Thus it is more likely to be related to the complexity aspect of 10-K filings and cause an increase in investor perceived uncertainty. In contrast, when risk factors

[14] When it is costly to access and process the information contained in the larger and more complex 10-K reports, the higher acquisition cost may crowd out informed investors while mitigating adverse selections (Goldstein and Yang (2017)), and thus improve market liquidity. We present empirical evidence of this crowding out effect using analyst forecast and institutional trading data in the online appendix.

[15] As the section titles already indicate the topics of discussions, this step of extracting individual sections alone can be viewed as a topic analysis. We also analyze another interesting section of 10-K filings on Legal Proceedings and present the results and discussions in the online appendix.

[16] See the SEC note (https://www.sec.gov/fast-answers/answersreada10khtm.html) on how to read a 10-K and how the risks might be presented in Risk Factors and MD&A sections. Some other studies also analyze the content of risk factors in various settings, such as the banking sector (Hanley and Hoberg (2019)), debt IPOs (Cai, Hanley, Huang, and Zhao (2020)), RMBS (Zhang, Zhao, and Zhao (2020)), and equity returns (Israelsen (2014); Lopez-Lira (2020)).



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are discussed in the MD&A section, the management may give their views of the risk factors and how the company is going to address them. In addition, the MD&A section also includes discussions on conditions and operational results and the management's perspective of the results. Therefore the MD&A section is more likely to reveal the information content aspect of 10-K filings and help resolve uncertainty. The second step is to employ a machine learning topic analysis algorithm called Non-Negative Matrix Factorization (NMF) to decompose each section to individual topics and quantify the extent of discussions in every topic. After we construct the measures, we validate them by showing their association with 10-K file size and a normalized measure of hard information. We then show that the newly constructed measures affect short run and long run volatility dynamics separately in Section 3.1. In addition, we construct two alternative measures of information content and analyze how they affect volatility dynamics in Section 3.2 and Section 3.3.

### 3.1 Topic analysis and the measures of information content and complexity

NMF is a static topic modeling algorithm that aims to automatically discover the hidden thematic structure in a large corpus of text documents and describe data clusters of related documents. The basic intuition of NMF is to decompose a document-term matrix V into the product of two non-negative matrices W and H, where W is an N×T document-topic matrix and H is an T×M topic-term matrix (N = the number of documents, M = the number of unique words, T = the number of topics).

We apply the NMF algorithm for the individual section (Risk Factors and MD&A separately) using all the 10-K filings. Consistent with the literature, we find that the individual section can be well summarized by 5 interpretable topics.[17] For both Risk Factors and MD&A, we score each section's exposure to their own 5 topics. These topics are summarized in the online appendix. The NMF analysis can be viewed as a decomposition of a section of disclosure into smaller pieces belonging to different topics (e.g., Lee and Seung (1999)), and the topic scores capture both the relevance of a section to a topic as well as the extent of discussions on that topic, consistent with our main goal to quantify the amount of information content in a topic. In contrast, another popular topic analysis algorithm Latent Dirichlet Allocation (LDA) is a probabilistic (as opposed

[17] In our topic analysis, we start with ten topics and conduct subsequent analysis based on the five most interpretable topics. However, our main results in Table 3 are robust using all ten topics.



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to static) model and the topic scores from LDA represent the probabilities of a document belonging to individual topics. Therefore, the topic scores from LDA are normalized (and sum up to 1) and can only proxy for the relevance of a section to a topic but does not capture the amount of disclosure on that topic. Moreover, prior research has suggested that NMF provides more coherent topics than LDA when the textual corpus has a more concentrated vocabulary (e.g., O'Callaghan, Greene, Carthy, and Cunningham (2015)). Thus NMF suits our purpose of decomposing the 10-K sections better.

After constructing the individual topic scores using NMF, we construct the aggregate (section) topic score by summing up a 10-K's individual topic scores for each section and use the aggregate topic score as a measure of disclosure amount/length in Risk Factors and MD&A, respectively. Before delving into the details of individual topics in each section, we first examine the relation between aggregate topic score and 10-K file size and the implication on the volatility dynamics.[18]

We regress the 10-K file size (Fsize) on the aggregate topic score controlling for the other variables used in the paper. The results are presented in column (1) of Panels A and B in Table 3. We find that both of the aggregate topic scores in the Risk Factors and MD&A are positively related to 10-K file size, suggesting they are both important components of the 10-K file size.

Next, we regress the volatility dynamics in the short-run (diffs), in the long-run (diffl), and the difference between diffs and diffl (diffsl), on the each of the aggregate topic scores, controlling for other variables. The results are presented in columns (3) to (5) of the panels in the table. In contrast to the relation with file size, we find opposite effects of these two components on the volatility dynamics. The disclosure amount (proxied by the aggregate topic score) in the MD&A section is associated with significant reduction in long-run volatility dynamics (but insignificant relation with short-run dynamics), while the disclosure amount in the Risk Factors section is associated with significant increase in short-run volatility dynamics (but insignificant relation with long-run dynamics). This sharp contrast highlights that Risk Factors and MD&A capture the complexity and information content aspects of 10-K filings separately.

To sharpen the identification of the mechanism for the topic measures, we also regress the

[18]We obtain the textual data on the individual sections from SeekEdgar. Note that the Risk Factors on Item 1A of the 10-K filings became a mandated disclosure item after the Securities Offering Reform became effective on December 1, 2005 (see page 257 of the Securities Offering Reform final rule: https://www.sec.gov/rules/final/33-8591.pdf). Therefore the section data of Risk Factors (on Item 1A) are available after December 1, 2005 from SeekEdgar and our topic analysis is limited to the sample from this date onward.



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percentage of numbers in 10-Ks (a normalized proxy for hard information) on the aggregate topic scores. The results are reported in column (2) of the panels in the table. We find that the complexity component (through Risk Factors) is negatively and statistically significantly associated with the normalized measure of hard information (percentage of numbers). In contrast, the information content component (through MD&A) is positively and statistically significantly associated with the measure of hard information. These analyses provide further validation of the text-based measures that separately capture the complexity and information content of 10-K filings. Intuitively, the discussions of topics in Risk Factors are text heavy and thus are perceived by investors with higher uncertainty in the future (in line with short-run increase in the volatility dynamics) whereas discussion of topics in MD&A carries more information content with delays in digesting the information (in line with long-run reduction in the volatility dynamics).

[Insert Table 3 here]

In the online appendix, we report the details of the procedures for NMF and repeat the same regression analysis for each individual topic in Risk Factors and MD&A. We find that for MD&A, all the individual topic scores are positively associated with the 10-K file size, negatively associated with long-run volatility dynamics, and positively associated with the percentage of numbers. The topics cover various aspects of financial and operational results. For Risk Factors, three of the individual topic scores are positively associated with the 10-K file size, positively associated with short-run volatility dynamics, and all are negatively associated with percentage of numbers. These results suggest that MD&A is a more robust measure of information content than Risk Factors as a measure of complexity.

In the online appendix, we also present the results based on the Legal Proceedings section of 10-K filings. We find that the aggregate topic score of this section is positively associated with 10-K file size but does not impact the volatility dynamics in a significant way. This suggests that not all disclosures are equally informative, and highlights the importance of understanding the disclosure content when using disclosure length as a proxy for complexity or information content.

Despite the encouraging results from our decomposition analysis, we realize that it is impossible to completely separate the information content aspect and complexity aspect of 10-K filings. As the different sections of 10-K filings are interrelated, it is still meaningful and interesting to study



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their joint effects reflected in 10-K file sizes along the time dimension.

### 3.2 Normalized measure of information content and complexity

In the previous subsection we use the percentage of numbers in 10-K filings as a normalized measure of hard information. We now examine how the volatility dynamics are impacted by this normalized measure of information content and normalized measures of complexity (such as Fog) and how 10-K file size is associated with these measures.

We regress the 10-K file size (Fsize), the volatility changes in the short-run (diffs), in the long-run (diffl), and the difference between diffs and diffl (diffsl), on the percentage of numbers and three normalized measures of complexity (Fog, Flesch, and Kincaid index), controlling for other variables used in the paper.[19] The results are presented in Table 4. Panel A reports the results based on percentage of numbers (Pnum). We find that larger 10-K is associated with higher percentage of numbers in the filing. The percentage of numbers is negatively associated with volatility changes in both the short and long run, with stronger effect in the long run. This suggests that for 10-K filings with a higher percentage of numbers, the volatility does not increase as much as for the 10-K filings with a lower percentage of numbers in the short run. In the long run, the volatility decreases more for 10-Ks with higher Pnum. The evidence suggests that Pnum is an effective measure of information content in 10-K filings.

Panel B of Table 4 reports the results based on the Fog index. We find that the Fog index is positively associated with the 10-K file size. In the short run, a higher Fog index corresponds to more increase in the short-run volatility. In the long run, the volatility dynamics is also positively associated with the Fog index though the association is not statistically significant.[20] Panel C and D report the results based on two other normalized proxies for complexity (Flesch index and Kincaid index). The results are generally consistent with the results based on the Fog index.[21]

[19] All these three normalized complexity measures are based on two out of three characteristics of 10-K filings: percentage of complex words, words per sentence, and syllabus per word. Note that a higher Flesch index indicates that the text is earlier to read, so the correlations of Flesch index with Fog index and Kincaid index are negative.

[20] The results on Fog do not conflict with our main findings that, on average, long run volatility is lower compared to the volatility in the filing week. The positive coefficient between Fog and long-run volatility changes suggests that the reduction in long-run volatility is still less for firms with higher Fog index, although the cross-sectional difference in diffl between high Fog and low Fog firms is not as pronounced as in the short run for diffs.

[21] In the online appendix, we also report the results based on another normalized measure of readability, Common Words, and we find the results based on Common Words are similar to the results based on the other normalized measures of readability/complexity.



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[Insert Table 4 here]

Our findings suggest that the normalized measures of information content and complexity also capture these two aspects in the 10-K filings separately, adding further support on the dual roles of information in 10-K filings.

### 3.3 A measure of "new" 10-K information content

An alternative way of extracting the informative component in a textual document is through the comparison of the word usage. We apply this approach to 10-Ks. The assumption is that when the word usage changes in 10-Ks from one year to another, there is likely to be new information disclosed. However, this approach does not account for the topics or the structure of discussions. Although it might filter out the boilerplate text, this approach would not be able to extract the new information discussed with similar word usage. Thus we view the measure constructed using this approach as a noisy proxy of "new" information in 10-Ks.

Specifically, we follow recent papers (e.g., Hanley and Hoberg (2010), Hanley and Hoberg (2012), and Cohen, Malloy, and Nguyen (2020)) and construct our measure of information content based on the COSINE similarity of 10-K filings from consecutive years. If a firm $i$'s 10-K filing in year $t$ can be represented by the word frequency vector

$$W_{i,t} = [f(word_1), f(word_2), ..., f(word_n)],$$

where $f(word_j)$ denotes the frequency of $word_j$ in the 10-K filing, the COSINE similarity between firm $i'$s 10-Ks in year $t$ and $t-1$ can be calculated as

$$\text{COSINE}_{i,t} = \frac{W_{i,t} * W_{i,t-1}}{\|W_{i,t}\| \times \|W_{i,t-1}\|},$$

where $*$ represents the dot product, and $\|\|$ represents the Euclidean norm. When two 10-Ks have no overlap, the COSINE similarity is zero, whereas two identical 10-Ks have a COSINE similarity of 1, so $(1 - \text{COSINE}_{i,t})$ captures the percentage of "new" information in firm $i$'s 10-K in year $t$. We multiply $(1 - \text{COSINE}_{i,t})$ by the raw file size in year $t$ to obtain the file size related to "new" information content, and the log of the new file size is denoted as New-Fsize. We use New-Fsize as



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a measure of "new" information content.[22]

We regress the volatility dynamics on the New-Fsize and the control variables. Table 5 reports the results from these regressions. Columns (1) and (2) of Table 5 show that the impact of New-Fsize on the short-run changes in volatility (diffs) is weaker than that in Table 1. Without controlling for firm-specific and disclosure-specific variables in column (1), the coefficient on New-Fsize is positive and statistically significant. However, when we include these control variables in column (2), the New-Fsize coefficient becomes negative though insignificant. The weaker results in columns (1) and (2) suggest that New-Fsize is effective in isolating the complexity aspect of 10-K filings from its information content. Moreover, the result in columns (3) and (4) shows that New-Fsize has a strong correlation with the reduction in volatility in the long-run (diffl). These findings indicate that, the New-Fsize well captures the information content aspect of 10-K filings that helps to resolve uncertainty in the longer horizon.

[Insert Table 5 here]

# 4 Additional results and robustness analyses

In this section, we present some additional results and various robustness analyses for the main findings. We organize the results in four parts. In section 4.1, we discuss the economic magnitude of our main findings using an option trading strategy. In section 4.2, we analyze how our main findings are affected by the transparency of information environment measured by analyst coverage and institutional ownership. In section 4.3, we control for several variables that could be related to confounding information sources in our main results. In section 4.4, we summarize how we address the impact from the subsequent earnings announcement that follows the 10-K filings.[23]

[22] In the online appendix, we find that the market reactions are positively and significantly associated with New-Fsize using three different measures of market reactions.

[23] In the online appendix, we provide the empirical evidence of crowing-out effect using analyst forecast and institutional trading data in Section OA.5.1, study how the market reacts to 10-K filings with new information in Section OA.5.2, investigate the possibility of a reverse causality in Section OA.5.3, address concerns regarding the contaminating effect of additional information production by intermediaries in Section OA.5.4, further investigate the volatility dynamics around earnings announcements in Section OA.5.5, perform option trading strategy based on counterfactual 10-K filing date in Section OA.5.6, and implement the butterfly option strategy in Section OA.5.7.



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### 4.1 The economic magnitude: an option-based investment strategy

The economic magnitude of our main findings can be compared with Loughran and McDonald (2014) and Pan, Wang, and Weisbach (2014). In Loughran and McDonald (2014), a one-standard-deviation increase in 10-K file size leads to about 4% of subsequent volatility's standard deviation. In our analyses, a one-standard deviation increase in 10-K file size is associated with a 3.6%-standard-deviation (0.248*1.265/8.8) increase in the subsequent short-horizon volatility (diffs), 4.4%-standard-deviation (0.487*1.265/13.9) decrease in the subsequent long-run volatility (diffl), and 8.3%-standard-deviation (0.752*1.265/11.5) change from the short-run to long-run (diffsl).[24] Therefore, while the predictive power of 10-K files size in the short-horizon and long-horizon volatility changes are comparable to that in Loughran and McDonald (2014), 10-K file sizes have more than twice the explanatory power for the volatility change from short run to longer horizon (diffsl). Similarly, Pan, Wang, and Weisbach (2014) find that on average the learning of CEO ability leads to a reduction of implied volatility level from around 0.175 to 0.155 in a 36-month horizon, representing a reduction of 11% of the initial implied volatility level in this time period (Figure 1 of their paper), or 0.44% per 6-week. In our analyses, a one-standard-deviation increase in 10-K files size leads to a 0.95% decrease in volatility in the 6 weeks from short run to long run.

From a theoretical point of view, the options price (both call and put) in the Black-Scholes model has a smooth, monotonically increasing, and convex relation with the underlying asset return volatility. Given that the implied volatility at the firm level is highly persistent, options price can be locally approximated as a linear function of volatility. Therefore, predicting volatility changes can also be considered as predicting option returns. To explore the option return equivalence of the volatility pattern documented in the previous sections, we use a strategy based on a short position of a straddle between two-to-four weeks (short-run) and eight-to-ten weeks (long-run) after 10-K filings. A straddle is an options strategy in which an investor holds a position in both a call and a put with the same strike price and expiration date.[25] A long position in a straddle is betting on an increase in future volatility, and a short position is betting on a decrease. Existing literature has documented that writing a straddle unconditionally creates economically and statistically significant

[24] In our sample, the standard deviation of diffs, diffl, and diffsl is 8.8(%), 13.9, and 11.5, respectively.

[25] For the rest of this study, we only report the results for the simple straddle, that is, a pair of call and put options with matching strike prices and maturity dates, due to its simplicity. We also tried the delta-neutral straddle and the result is very similar.



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profits, even after taking into account the transaction costs. In the context of our setting, the return on such a trade for each 10-K filing is equal to:

$$\text{Ret} = \frac{\text{Price(short-run)}}{\text{Price(long-run)}} - 1.$$

Panel A of Table 6 reports the average returns of our strategy for firms sorted into quintiles by previous year's 10-K file size. We use two methods to implement the strategy. The first method averages the straddle options prices for the days within weeks 2–4 and for the days within weeks 8–10, and the return is calculated as the percentage difference in the average prices. This implementation is equivalent to a price-weighted strategy (PW). The second strategy assumes a holding period of exactly six weeks (42 days). For example, if a straddle is written on the 21st day after a 10-K filing, the position is closed on the 63rd day after the filing. We then take the average of the returns across all days within the investment windows. This corresponds to an equal-weighted strategy (EW). As reported in Panel A of Table 6, our strategy generates large profits conditional on 10-K filings. The average 6-week PW return increases from 6.16% for firms with small 10-Ks to 8.05% for firms with large 10-Ks. The difference (1.89%) is statistically significant and corresponds to around 16.4% per year. Similar results hold for the EW strategy.

[Insert Table 6 here]

To gauge risk and return tradeoff of the investment strategies in Panel A, we report their annualized Sharpe ratio in Panel B. Since the strategies are event-based, we are unable to create portfolios with continuous and homogenous monthly returns. Therefore, we aggregate the option strategy returns for all 10-K disclosure events within a year (defined by 10-K filing dates) for each of the Fsize portfolios, construct the annual time series portfolio returns, and then compute the annualized Sharpe ratio of the long-short portfolio. Panel B shows that the annualized Sharpe ratios are 1.55 and 1.40 for the price-weighted strategy and the equally-weighted strategy, respectively. These Sharpe ratios are economically large compared with conventional equity strategies, such as market, size, value, and momentum.

To further examine the effect of firm characteristics including size, book-to-market ratio, momentum, gross profitability, asset growth, and idiosyncratic volatility, we run panel regressions of



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option strategy returns on 10-K file sizes with or without controlling these lagged firm characteristics. Since Fsize and firm size have an upward trend over time, we demean Fsize and firm size by their average values during the previous year. The results are reported in Panel C. For the PW strategy, the coefficient on Fsize is 1.33 with a standard error of 0.28 in the univariate regression. The coefficient is reduced to 0.7 but remains statistically significant when we control for these firm characteristics. The results are similar for the EW strategy. Therefore, the profitability of the investment strategies in Panel A cannot be explained by the standard firm characteristics.

In the online appendix, therefore, we implement the butterfly strategies that limit the downside losses. We find butterfly is similar but slightly more profitable than short straddles in Panel A, which confirms that our option trading profits are not driven by the exposure to the unlimited downside risk.

### 4.2 Analyst coverage, institutional ownership, and the volatility dynamics

In this subsection we examine how the impact of the 10-K filings on the volatility dynamics varies with firms' information environment. We proxy for the transparency of information environment using analyst coverage and institutional ownership. After splitting the observations into sub-samples with high and low level of analyst coverage (or institutional ownership), we run our main regressions within each sub-sample.

We report the results in Table 7. Panel A and B are based on analyst coverage and institutional ownership, respectively. We find that the increase in the short-run volatility is stronger for firms with low analyst coverage and low institutional ownership. This is consistent with the intuition that less transparent information environment impedes information incorporation by investors and correspondingly investors perceive higher uncertainty overall in the short-run. In contrast, more transparent information environment (more analyst coverage and institutional ownership) facilitates information processing and transmission in the long-run. Therefore, the incorporation of the 10-K information is relatively more thorough and subsequent uncertainty resolution is more complete by the investors. Indeed, we find that firms with high analyst coverage and high institutional ownership experience more reduction in the long-run volatility corresponding to a larger 10-K.

[Insert Table 7 here]



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### 4.3 Controlling for confounding sources of information

Our research design is in an event study setting and hence confounding information is always an important concern. It could either drive the empirical findings or contaminate the results.

We first control for the effect of other corporate disclosures that could be trigged by 10-K filings. In this case, a larger 10-K filing might trigger more follow-up disclosures which could generate the volatility patterns in our results. We use 8-K reports to control for these unscheduled material information releases (e.g., Zhao (2017)). We do so by including the frequency of 8-Ks in our analysis. Specifically, when explaining the short-run volatility changes, we include the frequency of 8-Ks in weeks two to four (denoted as #8K-s). Similarly, we include the 8-K frequency in weeks eight to ten (#8K-l) when analyzing the long-run volatility changes and we include #8K-sl(=#8K-s−#8K-l) when analyzing diffsl. Panel A of Table 8 shows that our main findings remain robust to controlling for the 8-K disclosures. [26]

Besides the firm's own information disclosure, another important driving factor of the volatility dynamics could be the additional information production by information intermediaries based on 10-K information content. For example, institutional investors might actively acquire more information after the 10-K filings and this could help reduce the long-run volatility. Ruling out all the alternative information sources in an ad-hoc, one-by-one fashion is virtually impossible. To this end, we focus on a database with comprehensive sources of information, the Ravenpack news database. Ravenpack covers news articles related to individual companies from reputable content sources, including Dow Jones Newswires, the Wall Street Journal and over 20000 other traditional and social media sites. It is widely used by institutional investors.

Similar to how we construct the 8-K frequency, we count the (natural logarithm of) total number of news articles on a firm in the Ravenpack database in three different horizons that correspond to diffs, diffl, and diffsl and we name the variables News-s, News-l, and News-sl, respectively. With the news measures, we conduct two sets of analyses. In the online appendix, we report the results to show that longer 10-Ks are indeed associated with more news production about the firm. In Panel B of Table 8, we report the results on the impact of 10-K file size on volatility dynamics after

[26] It is also possible that the management become less careful about the readability of 10-Ks when they are more comfortable about the future performance of the firm and expect lower uncertainty, which might cause the negative relation between 10-K file size and long-run volatility dynamics. Using the realized future fundamental information as a proxy for the management's information set, we address this potential reverse causality issue in the online appendix.



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controlling for the counts of total number of news articles. Our results show that the frequency of other news articles does not affect our main results. However, the caveat of this approach is that we can only control for the private information production to the extent of their correlation with all the public sources of information.

In addition, we address the impact of analyst specific news by focusing on the news that are related to the initiation, revision, and reviewing of analyst ratings and forecasts. We report the results in Panel C of the table. After controlling for the analysts news (News-Analyst-s, News-Analyst-l, News-Analyst-sl), we find that the impact of 10-K file size on the volatility dynamics remains robust. These results together suggest that 10-K filings do provide valuable information to the information intermediaries but our main result on the information content role of the 10-Ks is not likely to be mainly driven by the information produced by the information intermediaries.

Another related concern is that 10-K file size may be highly correlated with firm's structural complexity (e.g., Cohen and Lou (2012)). A firm with complex business structure is likely to file large 10-K reports to describe its operations, and our main result may be driven by this firm complexity. To address this concern, we follow LMD and repeat our main regressions by controlling for a firm complexity measure – the number of business segments from Compustat Segment Data. As reported in Panel D of Table 8, the magnitude of the coefficient on the 10-K file size slightly weakens in the short run, but remains almost the same in the long run.

[Insert Table 8 here]

### 4.4 Addressing the impact of earnings announcements

Our last set of robustness analyses addresses the impact of the subsequent earnings announcements. One concern about our benchmark sample is that the subsequent quarterly earnings announcements might be too close to the 10-K filing dates. The anticipation of subsequent earnings announcements may contaminate the 10-K information reflected in the volatility dynamics following 10-K filings.

Figure 3 shows that uncertainty is high before earnings announcements but drops immediately after their release. Therefore the hump-shaped volatility dynamics may be partly affected by the impact of the subsequent earnings announcement. In addition, if the two aspects of 10-K filings are indeed important, one could expect our main result to be even stronger by excluding firms whose



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gap between the 10-K filing and the subsequent earnings announcement is too short.

[Insert Figure 3 here]

To alleviate the impact of the confounding information due to quarterly earnings announcements, we repeat our analysis in the cross-section using only firms with no earnings announcements in at least seven weeks after 10-K filings.[27] If our main result in the cross section were driven by the subsequent earnings announcements, the coefficients on 10-K file size in the panel regressions using this subsample should be much weaker.

Table 9 presents the results. We find that the impact of 10-K file size on the short-run volatility dynamics does not change much whereas the impact on the long-run volatility dynamics becomes much stronger. Ceteris paribus, now a one standard deviation increase in log file size is associated with $0.193 \times 1.147 = 0.221$ percent increase in implied volatility in the short run, but is associated with $1.023 \times 1.147 = 1.173$ percent decrease in implied volatility in the relatively long-run following 10-K filings.[28] The marginal impact of the 10-K file size on the resolution of uncertainty almost doubles in the screened subsample, indicating that subsequent earnings announcements are unlikely to be the major source of the pattern in the implied volatility dynamics documented in Table 1.

[Insert Table 9 here]

We report two additional sets of results related to earnings announcements in the online appendix. First, we use simulations to construct a sample of *counterfactual* 10-K filings whose volatility dynamics are purely drawn from the empirical distribution of volatility dynamics around earnings announcements, and then we analyze the volatility dynamics for this simulated sample. We find that in this counterfactual sample, both the average and cross-sectional volatility dynamics patterns are different from our main findings. In the second set of analyses, we address the concern that managers may use large-size 10-Ks to obfuscate information when earnings are bad and our main findings may only exist in the sub-sample with negative earnings surprises. We split our sample into the sub-samples of positive and negative earnings surprises and find that the volatil-

[27] In the pre-2006 sample, the median gap between 10-K filings and the subsequent earnings announcements is about seven weeks. Our choice of this cutoff time preserves roughly half of our earlier sample. However, our results are not driven by this particular choice and are robust to using longer cutoff times.

[28] The standard deviation of log file size is slightly different from that in the full sample (1.147 vs. 1.265).



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ity dynamics pattern exists for both sub-samples, although the effect is slightly stronger in the sub-sample of negative earnings surprises.

## 5 Conclusion

In this paper, we evaluate the impact of complexity and information content of the 10-K fillings on the evolution of uncertainty dynamics following the fillings. We find a hump-shaped volatility dynamics for an average firm and that this hump-shape is more pronounced for firms with larger 10-Ks. We provide an economically interpretable decomposition of the widely used 10-K file size measure, and create measures of information content that can be separated from the complexity aspect. Our content analysis of the 10-K filings also suggests that not all disclosure is equally informative, and it is important to understand the disclosure content when using disclosure length as a proxy for information content or complexity.

Our central findings suggest that, consistent with the fundamental goal of annual reports, a larger 10-K report carries more information content that dominates in the long run, although the complexity aspect of such signal can lead to a higher uncertainty in the short run, before the information is fully processed by investors. Our analysis highlights the importance of learning through time for understanding the opposing effects of complexity and information content on uncertainty dynamics. These findings thus provide useful guidance for theoretical research on learning and asset pricing, especially in settings where the informative signal is complex and difficult to process. Our findings also have important policy implications on disclosure regulation. The disclosure regulation can focus on summarizing key information that is easier for investors to incorporate more quickly.[29] The recent disclosure regulation change on the requirement of Inline XBRL started in 2018 should help investors achieve this goal.

[29]For example, Sheng, Xu, and Zheng (2021) show that the summary prospectus of mutual funds (adopted in 2009) enhances the informativeness of disclosure to investors.



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**Table 1: The volatility dynamics following 10-K filings and the impact of 10-K file size**
This table reports the dynamics of option implied volatility in the (relatively) short and long horizons following 10-K disclosures. All the volatility changes are relative to filing-week volatility level; diffs is the average percentage change in the implied volatility 2-4 weeks after the 10-K disclosure; diffl is the average percentage change in the implied volatility 8-10 weeks after the 10-K disclosure; diffsl is equal to diffs minus diffl. Panel A presents the mean of diffs, diffl, and diffsl, and the robust standard errors clustered at the firm level are reported in parentheses below each coefficient estimate. Panel B reports the effect of 10-K file size (Fsize) from Loughran and McDonald (2014) on the dynamics of option implied volatility in the three different horizons (diffs, diffl, and diffsl) following 10-K disclosures. Other control variables include the logarithm of firm's market cap (Size), the logarithm of book-to-market equity ratio (BM), the logarithm of the average implied volatility [-252,-6] days before 10-K disclosure (Pre-IV), the CAPM alpha [-252,-6] days before 10-K disclosure (Pre-alpha), the absolute value of filing period excess return (Abs-filing-ret), and calendar year dummy variables (not reported). The robust standard errors clustered at the firm level are reported in parentheses below each coefficient estimate. The sample period is 1996-2012. Statistical significance levels of 1%, 5%, and 10% are indicated with ***, **, and * respectively.

Panel A: Average volatility dynamics

| | (1) diffs | (2) diffl | (3) diffsl |
|---|---|---|---|
| Mean | 0.360*** | -2.211*** | 2.569*** |
| | (0.06) | (0.10) | (0.08) |

Panel B: Impact of Fsize on volatility dynamics

| Dependent variable | (1) diffs | (2) diffs | (3) diffl | (4) diffl | (5) diffsl | (6) diffsl |
|---|---|---|---|---|---|---|
| Fsize | 0.500*** | 0.248*** | -0.283** | -0.487*** | 0.797*** | 0.752*** |
| | (0.08) | (0.09) | (0.13) | (0.15) | (0.11) | (0.13) |
| Size | | 0.296*** | | -0.071 | | 0.366*** |
| | | (0.05) | | (0.09) | | (0.08) |
| BM | | -0.153* | | -0.203 | | 0.038 |
| | | (0.08) | | (0.13) | | (0.11) |
| Pre-IV | | -3.105*** | | -7.650*** | | 4.145*** |
| | | (0.52) | | (0.83) | | (0.71) |
| Pre-Alpha | | 2.788*** | | 6.585*** | | -3.920*** |
| | | (0.33) | | (0.50) | | (0.42) |
| Abs-filing-ret | | 0.153*** | | 0.088*** | | 0.053** |
| | | (0.02) | | (0.03) | | (0.03) |
| N | 21,848 | 18,202 | 21,268 | 17,797 | 19,952 | 17,031 |
| R-sq | 0.082 | 0.105 | 0.124 | 0.148 | 0.166 | 0.182 |



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**Table 2: Impact of 10-K file size on alternative measures of uncertainty dynamics**

This table reports the effect of 10-K file size on alternative measures of uncertainty dynamics. The first alternative uncertainty measure is based on the firm-level ambiguity of the stock return distributions (Panel A), following Izhakian and Yermack (2017) and Augustin and Izhakian (2020). The second and third alternative measures are based on the information of analyst forecast: the analyst forecast dispersion (Panel B) and the absolute value of earnings surprises (Panel C). Both variables are scaled by stock price. The fourth is the residual root mean squared error (RMSE) from a market model estimated using daily observations (rolling 21 trading days) of individual firms' returns (Panel D) and 30-day options implied volatility (Panel E). We measure the percentage changes of these five uncertainty variables for the short run (diffs), long run (diffl), and the difference between short run and long run (diffsl) in the same way as we calculate our main volatility dynamics. The main independent variable of interest is the natural logarithm of the 10-K file size (Fsize). Other control variables are the same as in Table 1. The calendar year dummy variables are included in the regressions (not reported). The robust standard errors clustered at the firm level are reported in parentheses below each coefficient estimate. The sample period is 1996-2012. Statistical significance levels of 1%, 5%, and 10% are indicated with ***, **, and * respectively.

| | Panel A. Ambiguity | | | Panel B. Analyst Dispersion | | | Panel C. Abs(SUE) | | |
|---|---|---|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) |
| Dependent | diffs | diffl | diffsl | diffs | diffl | diffsl | diffs | diffl | diffsl |
| Fsize | 0.919*** | -0.758** | 1.255*** | 4.870*** | -3.264** | 4.515** | 6.592*** | -5.384** | 4.874* |
| | (0.31) | (0.36) | (0.37) | (1.24) | (1.36) | (1.76) | (1.57) | (2.11) | (2.52) |
| Control | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| N | 43,729 | 43,934 | 40,521 | 10,180 | 9,330 | 6,139 | 10,775 | 9,835 | 6,505 |
| R-sq | 0.063 | 0.188 | 0.063 | 0.016 | 0.023 | 0.025 | 0.009 | 0.010 | 0.020 |

| | Panel D. RMSE | | | Panel E. 30D options | | |
|---|---|---|---|---|---|---|
| | (10) | (11) | (12) | (13) | (14) | (15) |
| Dependent | diffs | diffl | diffsl | diffs | diffl | diffsl |
| Fsize | 0.846*** | -1.159*** | 1.662*** | 0.507*** | -0.512*** | 1.004*** |
| | (0.32) | (0.31) | (0.32) | (0.14) | (0.18) | (0.18) |
| Control | Yes | Yes | Yes | Yes | Yes | Yes |
| N | 61,586 | 59,693 | 59,165 | 29,191 | 28,288 | 28,279 |
| R-sq | 0.032 | 0.023 | 0.049 | 0.034 | 0.098 | 0.108 |



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**Table 3: Decomposition of Fsize to information content and complexity by topic analysis**

This table reports the effect of the aggregate topic score for Risk Factors (Panel A) and MD&A (Panel B) sections of 10-K filings on the following dependent variables: the natural logarithm of the 10-K file size (Fsize), the percentage of numbers (Pnum), the percentage changes of implied volatility dynamics in the short run (diffs), long run (diffl), and diffs minus diffl (diffsl). To construct the aggregate topic score, we first employ a machine learning topic analysis algorithm called Non-Negative Matrix Factorization (NMF) for the individual section (Risk Factors and MD&A separately) and extract 5 interpretable topics from each section. The aggregate topic score is measured by summing up a 10-K's individual topic scores for each section. More details of the topic analysis can be found in the online appendix to accompany this paper. All regressions include control variables described in Table 1. The robust standard errors clustered at the firm level are reported in parentheses below each coefficient estimate. The sections data are obtained from SeekEdgar. The sample period is limited to 2006-2012 due to the mandatory disclosure requirement on Item 1A (Risk Factors) on 10-Ks by Securities Offering Reform that became effective on December 1, 2005. Statistical significance levels of 1%, 5%, and 10% are indicated with ***, **, and * respectively.

Panel A: Topic Score: RiskFactors

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| Dependent variable | Fsize | Pnum | diffs | diffl | diffsl |
| TopicScore-RiskFactors | 0.049*** | -0.421*** | 0.223** | -0.081 | 0.244* |
| | (0.01) | (0.03) | (0.09) | (0.15) | (0.13) |
| Control | Yes | Yes | Yes | Yes | Yes |
| N | 6,430 | 6,382 | 5,703 | 5,594 | 5,388 |
| R-sq | 0.638 | 0.207 | 0.128 | 0.110 | 0.200 |

Panel B: Topic Score: MD&A

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| Dependent variable | Fsize | Pnum | diffs | diffl | diffsl |
| TopicScore-MD&A | 0.071*** | 0.229*** | 0.049 | -0.201** | 0.259*** |
| | (0.01) | (0.03) | (0.06) | (0.08) | (0.08) |
| Control | Yes | Yes | Yes | Yes | Yes |
| N | 6,443 | 6,395 | 5,714 | 5,606 | 5,399 |
| R-sq | 0.651 | 0.179 | 0.126 | 0.112 | 0.202 |



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**Table 4: Effect of normalized proxies for information content and complexity**

This table reports the effect of normalized proxies for 10-K information content and complexity on the implied volatility dynamics following 10-K fillings. The normalized proxy for 10-K information content is the percentage of numbers (Pnum) in the filing (Panel A). The normalized proxies for 10-K complexity are: the Fog index from Li (2008) (Panel B), the Flesch index (Panel C), and the Kincaid index (Panel D). The dependent variables are the natural logarithm of the 10-K file size (Fsize), the percentage changes in volatility dynamics in the short run (diffs), long run (diffl), and diffs minus diffl (diffsl). The main independent variables are Pnum (Panel A), Fog (Panel B), Flesch (Panel C), and Kincaid (Panel D). All regressions include control variables described in Table 1. The robust standard errors clustered at the firm level are reported in parentheses below each coefficient estimate. The sample period is 1996-2012. Statistical significance levels of 1%, 5%, and 10% are indicated with ***, **, and * respectively.

| | Panel A. Independent : Pnum | | | | Panel B. Independent : Fog | | | |
|---|---|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
| Dependent | fsize | diffs | diffl | diffsl | fsize | diffs | diffl | diffsl |
| Independent | 0.017*** | -0.072*** | -0.161*** | 0.094*** | 0.107*** | 0.178*** | 0.145* | 0.039 |
| | (0.00) | (0.02) | (0.03) | (0.03) | (0.01) | (0.05) | (0.08) | (0.07) |
| Control | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| N | 20,606 | 18,051 | 17,634 | 16,880 | 15,763 | 13,822 | 13,573 | 12,991 |
| R-sq | 0.740 | 0.104 | 0.154 | 0.190 | 0.696 | 0.099 | 0.151 | 0.174 |

| | Panel C. Independent : Flesch | | | | Panel D. Independent : Kincaid | | | |
|---|---|---|---|---|---|---|---|---|
| | (9) | (10) | (11) | (12) | (13) | (14) | (15) | (16) |
| Dependent | fsize | diffs | diffl | diffsl | fsize | diffs | diffl | diffsl |
| Independent | -0.015*** | -0.070*** | -0.058** | -0.013 | 0.120*** | 0.188*** | 0.202*** | -0.002 |
| | (0.00) | (0.02) | (0.03) | (0.02) | (0.01) | (0.05) | (0.07) | (0.07) |
| Control | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| N | 15,763 | 13,822 | 13,573 | 12,991 | 15,763 | 13,822 | 13,573 | 12,991 |
| R-sq | 0.679 | 0.099 | 0.151 | 0.174 | 0.701 | 0.099 | 0.152 | 0.174 |



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**Table 5: Effect of a measure of "new" 10-K information content on volatility dynamics**
This table reports the effect of a modified 10-K file size (New-Fsize) that captures "new" information content on the volatility dynamics following 10-K filings. The dependent variables are the percentage changes in volatility dynamics in the short run (diffs), long run (diffl), and diffs minus diffl (diffsl). The main independent variable of interest is the natural logarithm of the modified 10-K file size that captures "new" information content. To generate this measure for a firm in year $t$, we first calculate the COSINE similarity between this firm's 10-K in year $t$ and the 10-K in year $t-1$. Then we use $(1 - \text{COSINE}) \times \text{Size}_t$ to calculate the 10-K file size corresponding to the "new" information content. All regressions include control variables described in Table 1. The robust standard errors clustered at the firm level are reported in parentheses below each coefficient estimate. The sample period is 1996-2012. Statistical significance levels of 1%, 5%, and 10% are indicated with ***, **, and * respectively.

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| Dependent variable | diffs | diffs | diffl | diffl | diffsl | diffsl |
| New-Fsize | 0.294*** | -0.011 | -0.358** | -0.412** | 0.527*** | 0.371** |
| | (0.11) | (0.12) | (0.18) | (0.18) | (0.15) | (0.15) |
| Size | | 0.356*** | | -0.131 | | 0.490*** |
| | | (0.05) | | (0.08) | | (0.07) |
| BM | | -0.093 | | -0.249** | | 0.142 |
| | | (0.08) | | (0.13) | | (0.11) |
| Pre-IV | | -3.012*** | | -7.645*** | | 4.228*** |
| | | (0.53) | | (0.84) | | (0.71) |
| Pre-Alpha | | 2.654*** | | 6.571*** | | -4.063*** |
| | | (0.33) | | (0.51) | | (0.43) |
| Abs-filing-ret | | 0.158*** | | 0.080** | | 0.060** |
| | | (0.03) | | (0.03) | | (0.03) |
| N | 20,778 | 17,752 | 20,190 | 17,346 | 18,964 | 16,616 |
| R-sq | 0.082 | 0.107 | 0.127 | 0.150 | 0.170 | 0.184 |



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**Table 6: Economic impact of 10-K information content: An options strategy**

This table reports the potential profitability of an option investment strategy. Panel A reports the summary statistics in returns of a strategy with a short position in a straddle during 2 to 4 weeks after the 10-K filing that is closed in weeks 8 to 10. We report the results for both price-weighted (PW) and equal-weighted (EW) strategies. Panel B reports the annualized Sharpe ratios of the PW and EW strategies by combining strategy returns for all events within a calendar year for each Fsize quintile. Panel C reports the coefficient of Fsize in panel regressions in predicting straddle returns with or without controlling for firm characteristics including size, book-to-market, momentum, gross profitability, asset growth, and idiosyncratic volatility. To save space, we only report the coefficient of Fsize. To remove trend in Fsize and firm size in the panel regressions, we demean by their averages from the previous year. P10 and P90 represent the 10th and 90th percentiles for each variable. The difference in variables between the portfolio with large 10-K file size and the portfolio with small 10-K file size as well as the $t$-statistics from the $t$-test is reported in the bottom of each panel. The sample is from 1996 to 2012.

Panel A: Strategy returns for actual 10-K filings sample (%)

| | PW | | | | EW | | | |
|---|---|---|---|---|---|---|---|---|
| Fsize | Mean | Std | P10 | P90 | Mean | Std | P10 | P90 |
| Small | 6.16 | 28.88 | -30.86 | 40.06 | 8.16 | 22.56 | -20.36 | 30.93 |
| 2 | 6.12 | 28.40 | -30.34 | 39.99 | 7.61 | 22.27 | -21.09 | 29.76 |
| 3 | 5.93 | 29.20 | -30.96 | 39.85 | 7.53 | 21.89 | -20.52 | 29.08 |
| 4 | 6.69 | 29.05 | -29.71 | 40.67 | 8.40 | 22.68 | -19.01 | 28.92 |
| Large | 8.05 | 28.96 | -26.90 | 42.13 | 9.54 | 20.06 | -14.27 | 28.67 |
| Diff | 1.89 | | | | 1.38 | | | |
| $t$-stat | (6.32) | | | | (8.52) | | | |

Panel B: Annualized Sharpe ratio

| | PW | EW |
|---|---|---|
| SR | 1.55 | 1.40 |

Panel C: Short straddle return prediction by Fsize (%)

| | PW | | EW | |
|---|---|---|---|---|
| Fsize | 1.331*** | 0.700** | 1.176*** | 0.654** |
| | (0.28) | (0.27) | (0.27) | (0.26) |
| Ctrl | No | Yes | No | Yes |



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**Table 7: Effect of 10-K file size on volatility dynamics conditional on transparency of information environment**

This table reports the results on how the transparency of information environment affects the effect of 10-K file size on the volatility dynamics. We proxy for the information environment transparency using analyst coverage and institutional ownership. We split the sample to sub-samples of high and low analyst coverage in Panel A and high and low institutional ownership in Panel B. In each sub-sample, we regress the changes in volatility dynamics in the short run (diffs), in the long run (diffl), and diffs minus diffl (diffsl), on the logarithm of 10-K file size (Fsize). All regressions include control variables described in Table 1. The robust standard errors clustered at the firm level are reported in parentheses below each coefficient estimate. The sample period is 1996-2012. Statistical significance levels of 1%, 5%, and 10% are indicated with ***, **, and * respectively.

Panel A: Number of analysts

| | High | | | Low | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Dependent Variable | diffs | diffl | diffsl | diffs | diffl | diffsl |
| Fsize | 0.169 | -0.734*** | 0.895*** | 0.273** | -0.361 | 0.455** |
| | (0.12) | (0.19) | (0.16) | (0.13) | (0.22) | (0.20) |
| Control | Yes | Yes | Yes | Yes | Yes | Yes |
| N | 9,760 | 9,613 | 9,345 | 8,211 | 8,017 | 7,249 |
| R-sq | 0.137 | 0.181 | 0.213 | 0.095 | 0.137 | 0.156 |

Panel B: Institutional ownership

| | High | | | Low | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Dependent Variable | diffs | diffl | diffsl | diffs | diffl | diffsl |
| Fsize | 0.060 | -0.686*** | 0.679*** | 0.347*** | -0.226 | 0.545*** |
| | (0.12) | (0.22) | (0.19) | (0.12) | (0.19) | (0.17) |
| Control | Yes | Yes | Yes | Yes | Yes | Yes |
| N | 9,002 | 8,915 | 8,410 | 8,929 | 8,672 | 8,144 |
| R-sq | 0.116 | 0.154 | 0.168 | 0.117 | 0.170 | 0.204 |



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**Table 8: Controlling for confounding sources of information**

This table reports the results on the effect of 10-K file size on the volatility dynamics following 10-K filings, after controlling for several confounding sources of information. The dependent variables are the percentage changes in volatility dynamics in the short run (diffs), long run (diffl), and diffs minus diffl (diffsl). The main independent variable of interest is the natural logarithm of the 10-K file size (Fsize). In Panel A, we control for the frequency of 8-K filings (#8K-s, #8K-l, and #8K-sl) in the horizons that are consistent with the volatility dynamics (diffs, diffl, and diffsl). In Panel B, we control for the (natural logarithm of) total number of news articles in RavenPack database about the firm (News-s, News-l, and News-sl) in the horizons are consistent with the volatility dynamics (diffs, diffl, and diffsl). In Panel C, we control for the analyst specific (related to forecasts and ratings) news (News-Analyst-s, News-Analyst-l, News-Analyst-sl) in the horizons that are consistent with the volatility dynamics (diffs, diffl, and diffsl). In Panel D, we control for the number of business segments in the firm (# of Segment). All regressions include control variables described in Table 1. The robust standard errors clustered at the firm level are reported in parentheses below each coefficient estimate. The sample period is 1996-2012. Statistical significance levels of 1%, 5%, and 10% are indicated with ***, **, and * respectively.

Panel A. #8K filings

| | (1) | (2) | (3) |
|---|---|---|---|
| Dependent | diffs | diffl | diffsl |
| Fsize | 0.237*** | -0.509*** | 0.751*** |
| | (0.09) | (0.15) | (0.13) |
| #8K-s | 0.121* | | |
| | (0.07) | | |
| #8K-l | | 0.261*** | |
| | | (0.10) | |
| #8K-sl | | | 0.104 |
| | | | (0.07) |
| Control | Yes | Yes | Yes |
| N | 18,202 | 17,797 | 17,031 |
| R-sq | 0.105 | 0.148 | 0.182 |

Panel B. # of News

| | (4) | (5) | (6) |
|---|---|---|---|
| Dependent | diffs | diffl | diffsl |
| Fsize | 0.194** | -0.717*** | 0.889*** |
| | (0.10) | (0.16) | (0.13) |
| News-s | 0.127 | | |
| | (0.08) | | |
| News-l | | 0.246 | |
| | | (0.16) | |
| News-sl | | | 0.098 |
| | | | (0.11) |
| Control | Yes | Yes | Yes |
| N | 15,152 | 14,937 | 14,327 |
| R-sq | 0.122 | 0.163 | 0.205 |

Panel C. # of Analyst news

| | (7) | (8) | (9) |
|---|---|---|---|
| Dependent | diffs | diffl | diffsl |
| Fsize | 0.202** | -0.702*** | 0.891*** |
| | (0.09) | (0.16) | (0.13) |
| News-Analyst-s | 0.237* | | |
| | (0.14) | | |
| News-Analyst-l | | 0.485*** | |
| | | (0.18) | |
| News-Analyst-sl | | | -0.233 |
| | | | (0.19) |
| Control | Yes | Yes | Yes |
| N | 15,152 | 14,937 | 14,327 |
| R-sq | 0.122 | 0.163 | 0.205 |

Panel D. # of Segment

| | (10) | (11) | (12) |
|---|---|---|---|
| Dependent | diffs | diffl | diffsl |
| Fsize | 0.271*** | -0.421** | 0.732*** |
| | (0.10) | (0.16) | (0.14) |
| # of Segment | 0.001 | | |
| | (0.03) | | |
| # of Segment | | -0.174*** | |
| | | (0.05) | |
| # of Segment | | | 0.148*** |
| | | | (0.04) |
| Control | Yes | Yes | Yes |
| N | 15,173 | 14,891 | 14,227 |
| R-sq | 0.100 | 0.142 | 0.173 |



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**Table 9: Effect of 10-K file size on volatility dynamics: Analysis of sub-sample with no earnings announcements in at least 7 weeks after 10-K filings**

This table reports the effect of 10-K file size on the volatility dynamics following 10-K filings for the sub-sample of observations with no earnings announcement in at least 7 weeks after the filings. The dependent variables are the percentage changes in volatility dynamics in the short run (diffs), long run (diffl), and diffs minus diffl (diffsl). The main independent variable of interest is the natural logarithm of the 10-K file size (Fsize). Other control variables are described in Table 1. All regressions include calendar year dummy variables (not reported). The robust standard errors clustered at the firm level are reported in parentheses below each coefficient estimate. The sample period is 1996-2012. Statistical significance levels of 1%, 5%, and 10% are indicated with ***, **, and * respectively.

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| Dependent variable | diffs | diffs | diffl | diffl | diffsl | diffsl |
| Fsize | 0.816*** | 0.193* | -0.555*** | -1.023*** | 1.378*** | 1.207*** |
| | (0.10) | (0.12) | (0.15) | (0.19) | (0.13) | (0.17) |
| Size | | 0.431*** | | 0.001 | | 0.461*** |
| | | (0.08) | | (0.12) | | (0.10) |
| BM | | 0.115 | | -0.058 | | 0.163 |
| | | (0.11) | | (0.16) | | (0.14) |
| Pre-IV | | -5.050*** | | -9.273*** | | 3.988*** |
| | | (0.75) | | (1.13) | | (0.94) |
| Pre-Alpha | | 1.771*** | | 5.307*** | | -3.546*** |
| | | (0.50) | | (0.76) | | (0.62) |
| Abs-filing-ret | | 0.116*** | | 0.070 | | 0.052 |
| | | (0.03) | | (0.04) | | (0.03) |
| N | 12,268 | 10,669 | 12,061 | 10,518 | 11,401 | 10,119 |
| R-sq | 0.115 | 0.144 | 0.114 | 0.133 | 0.190 | 0.201 |



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**Figure 1: Average dynamics of implied volatility following 10-K disclosure**

This figure plots the dynamics of the option implied volatility following 10-K disclosures for an average firm. The percentage change of the implied volatility from the filing date is plotted against the number of weeks after the 10-K disclosure. The left panel plots the full sample 1996-2012, the middle panel plots the early sample 1996-2003, and the right panel plots late sample 2004-2012.

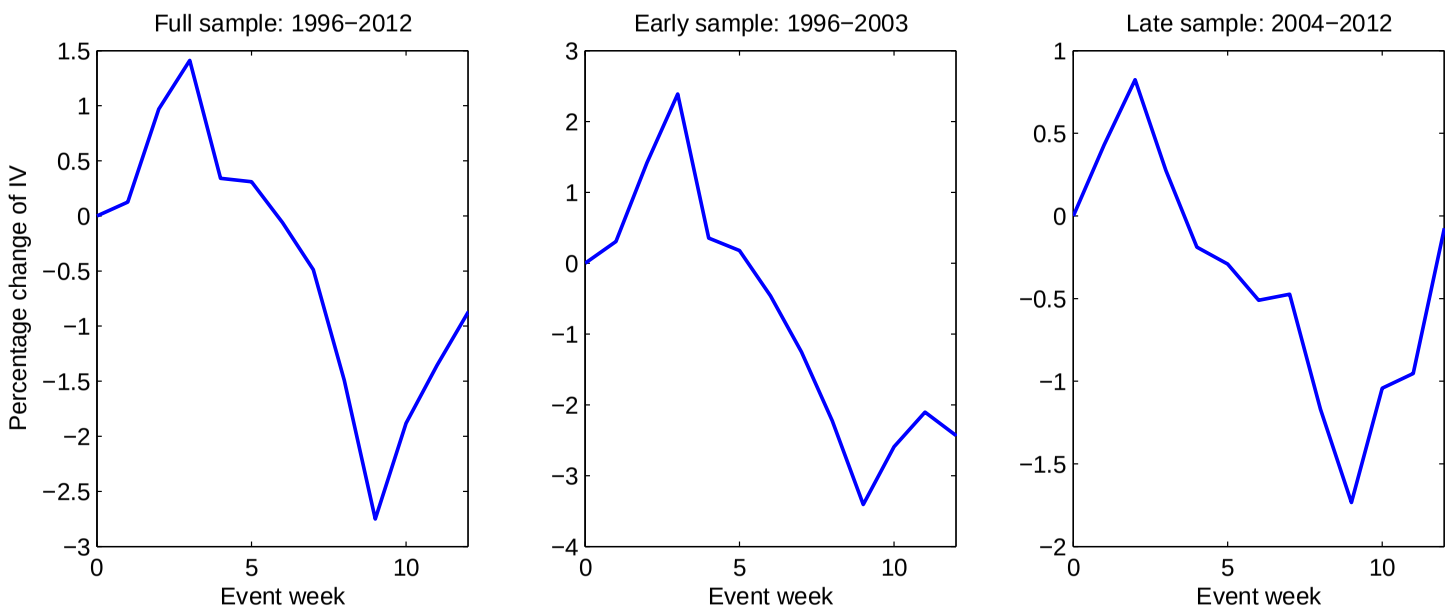




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**Figure 2: Dynamics of implied volatility for firms sorted by 10-K file size**

This figure plots the dynamics of the option implied volatility following 10-K disclosures for firms with small and large 10-K file size. In each year, firms are sorted into quintile portfolios based on their 10-K file size. The left panel plots the percentage change of the implied volatility from the filing date against the number of weeks after the 10-K disclosure for portfolio 1 (small 10-K file size) and portfolio 5 (large 10-K file size), whereas the difference is plotted in the right panel. The sample period is 1996-2012.

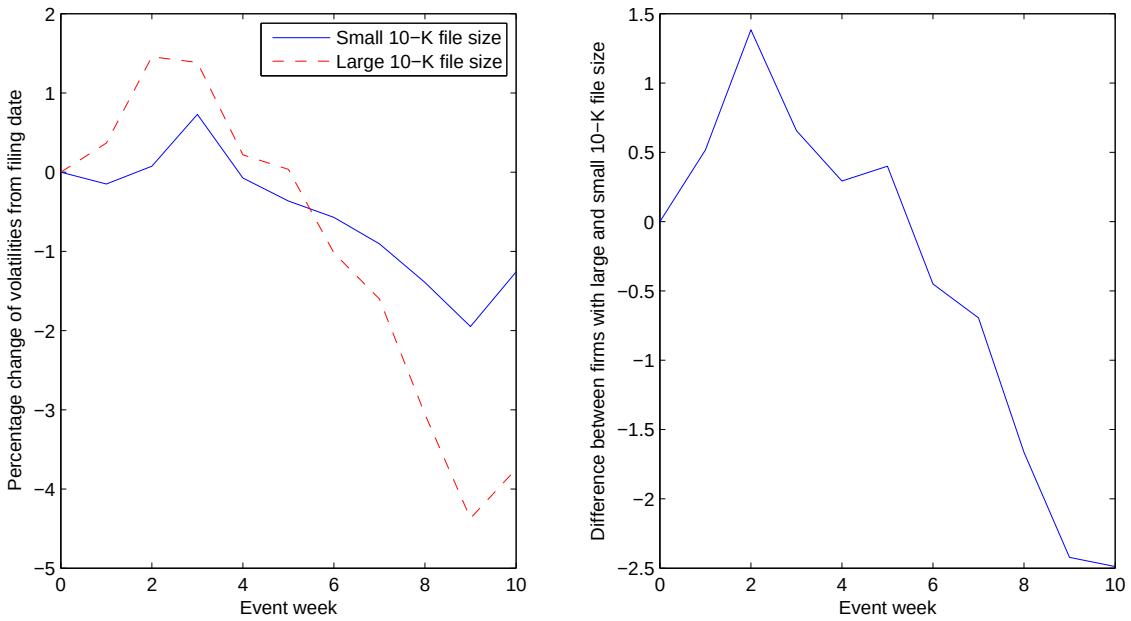




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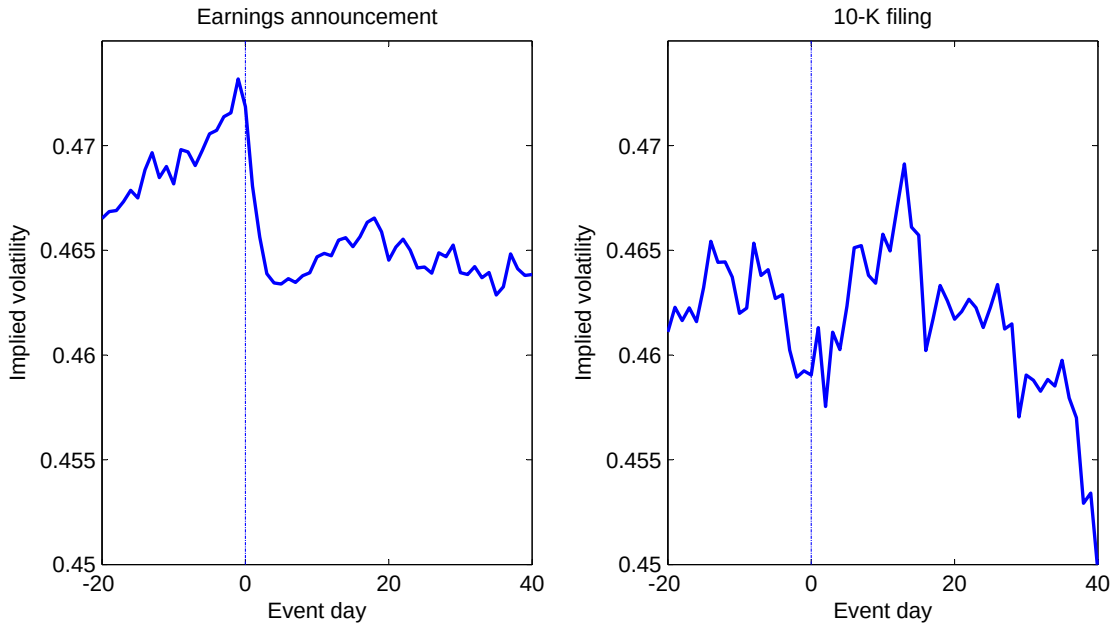
**Figure 3: Dynamics of implied volatility around earnings announcements and 10-K filings**

This figure compares the dynamics of the option implied volatility around earnings announcement (left panel) and 10-K disclosures (right panel). The option implied volatility is plotted against the number of trading days around these events. The sample is from 1996-2012.



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